concentration.[4] For example, the U.S. Department of Justice uses the HHI when examining potential mergers and acquisitions, to assess how they may influence the centralization of the industry.[5] The metric is calculated by taking the percent market share of each entity, squaring each market share, and summing these squares before multiplying by 10,000.

For example, a system that has two players with 50 percent market share apiece would have an HHI of 5,000, because $(0.5^2) + (.5^2) = 0.5$, and $0.5 \times 10{,}000 = 5{,}000$. For the HHI, anything less than 1,500 qualifies as a competitive marketplace, anything between 1,500 to 2,500 is a moderately concentrated marketplace, and anything greater than 2,500 is a highly concentrated marketplace.[6]

Blockchain networks should never classify as a highly concentrated marketplace, and ideally, should always fall into the competitive marketplace category. The more concentrated a marketplace is, the closer a single entity can be to gaining majority share of the compute power and performing a 51 percent attack. Figure 13.6 shows that both Bitcoin and Ethereum qualify as competitive marketplaces, while Litecoin is a moderately concentrated marketplace.[7]